

RILEY FRANCIS

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EDUCATION

University of Connecticut, Storrs, CT

Aug 2023 – Expected May 2027

Dual Degree: *BS Computer Science, BA Pure Mathematics*

Honors Program, Honors Scholar, COE Dean's List, CLAS Dean's List

GPA: 3.928 / 4.000

University of Edinburgh, Edinburgh, Scotland

Jan 2025 – May 2025

Visiting Student, School of Informatics

EXPERIENCE

SDE Co-op II

Apr 2023 – Present

Amazon Robotics, Westborough, MA

- Member of Amazon Robotics Innovation Lab, assisting in development of various robotic solutions.
- Developed ROS2 packages and CI tools for our lab.

Student Computer Programming Specialist

Apr 2023 – Jun 2025

University of Connecticut, Storrs, CT

- Handle redistribution and disposal of university electronic equipment.
- Configure and troubleshoot Windows, macOS, and Linux devices for reuse.

PROJECTS & RESEARCH

Reinforcement Learning on Autonomous Robots Inside Meta's Habitat-Lab

Feb 2021 – Jan 2023

Storrs, CT

Mentor: Dr. Jonathan Shihao Ji

- Developed a reinforcement learning method for training legged robots (i.e. [Spot](#)) inside Meta's Habitat-Lab.
- Collaborated in a team of four other researchers and formulated a research paper (available upon request).

Interpreting GNNs with MIPExplainer: Explaining Alzheimer's & Autism

May 2025 – Jul 2025

Greensboro, NC

Mentor: Dr. Chunjiang Zhu

- Applied MIPExplainer to brain GNNs to identify brain-region features influencing Alzheimer's and Autism prediction.
- Produced sparse model-level explanations to highlight key connectivity structures. Paper available upon request.

Making Lane Detection Efficient for Autonomous Model Cars

Feb 2021 – Jan 2023

Storrs, CT

Mentors: Dr. Jinbo Bi, Dr. Shengli Zhou, Dr. Caiwen Ding, Dr. Fei Miao

- Built and programmed autonomous model vehicles using AI and computer vision.
- Presented work at IEEE/ACM ISLPED.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, Java, HTML, CSS, JavaScript, TypeScript, React, LaTeX, SQL, PDDL

Packages: PyTorch, Torchvision, Tensorflow, Scikit-Learn, NetworkX, OpenCV, Gymnasium, NumPy, Matplotlib, Manim, SQLite, ROS2

Tools/Programs: Linux, Windows, GitHub, VS Code, Visual Studio, Unity, Obsidian, Google Suite, Office Suite

AWARDS & RECOGNITION

Honors Scholar (2023 – Present), CLAS Dean's List (2023 – Present), SoE Dean's List (2023 – Present), STEM Merit Scholarship (2023 – Present), 2025 Babbige Scholar (2025), Global Citizen & Merit Scholarship (Sep 2024)

ACTIVITIES

Science Olympiad at UConn, Treasurer & Co-Founder

Aug 2023 – May 2026

- Planned and coordinated invitational competitions for dozens of high school and middle school teams.

RESEARCH INTERESTS

Artificial Intelligence, Machine Learning, Computer Vision, Robotics, Graph Theory, Algorithms